# Differential Equations Workbook

Fourier series

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## FOURIER SERIES REPRESENTATIONS

- 1. Find the Fourier series representation of $f(x) = (2x + 1)^2$ on $-L \leq x \leq L$.

- 2. Find the Fourier series representation of $f(x) = e^x$ on $-\pi \leq x \leq \pi$.

- 3. Find the Fourier series representation of the function on $-\pi \leq x \leq \pi$.

$$f(x) = \sin\left(\frac{x}{2}\right)$$

- 4. Find the Fourier series representation of $f(x) = x \sin x$ on $-\pi \leq x \leq \pi$.

- 5. Find the Fourier series representation of $f(x) = x^2 + \sin x$ on $-L \leq x \leq L$.

- 6. Find the Fourier series representation of $f(x) = \cosh x$ on $-\pi \leq x \leq \pi$.

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## PERIODIC FUNCTIONS AND PERIODIC EXTENSIONS

■ 1. Sketch the function's periodic extension on the interval $-L \leq x \leq L$, given some positive value of $L$.

$$f(x) = \frac{1}{L}x^2 - L$$

■ 2. Sketch the function's periodic extension on the interval $-L \leq x \leq L$.

$$f(x) = \begin{cases} L + x & x < 0 \\ L - x & x \geq 0 \end{cases}$$

■ 3. Find the even and odd extensions of the function, given some positive value of $L$. Then sketch both extensions.

$$f(x) = \frac{1}{L^2}x^3 - \frac{1}{L}x^2$$

■ 4. Find the even and odd extensions of the function, given some positive value of $L$. Compare both sketches.

$$f(x) = \begin{cases} L^2 & \frac{L}{2} < x \\ 2Lx & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ -L^2 & x < -\frac{L}{2} \end{cases}$$

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■ 5. Find the even and odd extensions of the function, given some positive value of $L$.

$$f(x) = e^{L-2x}$$

■ 6. Sketch the periodic extensions of the even and odd extensions of the function $f(x) = x$ on $-L \leq x \leq L$.

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## REPRESENTING PIECEWISE FUNCTIONS

■ 1. Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} -2x & -L \leq x < 0 \\ 3x & 0 \leq x \leq L \end{cases}$$

■ 2. Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} 4 - 2x & -L \leq x < 0 \\ 2x + 4 & 0 \leq x \leq L \end{cases}$$

■ 3. Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} 1 & -L \leq x < 0 \\ 3\left(x + \frac{L}{2}\right) + 1 & 0 \leq x \leq L \end{cases}$$

■ 4. Find the Fourier series representation of the piecewise function on $-\pi \leq x \leq \pi$.

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$$f(x) = \begin{cases} (x+1)^2 & -\pi \leq x < 0 \\ 2-(x-1)^2 & 0 \leq x \leq \pi \end{cases}$$

■ 5. Find the Fourier series representation of the piecewise function.

$$f(x) = \begin{cases} 3x+6 & -L \leq x < -1 \\ -3x & -1 \leq x \leq 1 \\ 3x-6 & 1 < x \leq L \end{cases}$$

■ 6. Find the Fourier series representation of the piecewise function.

$$f(x) = \begin{cases} 2-2x & -L \leq x < 0 \\ 4x+3 & 0 \leq x \leq L \end{cases}$$

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## CONVERGENCE OF A FOURIER SERIES

■ 1. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} -1 & -L \le x < 0 \\ x^2 & 0 \le x \le L \end{cases}$$

$$f(x) = -\frac{1}{2} + \frac{L^2}{6} + \frac{2L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

$$+ \sum_{n=1}^{\infty} \frac{1}{n^3\pi^3} \left[(2L^2 - L^2n^2\pi^2 - n^2\pi^2)(-1)^n + n^2\pi^2 - 2L^2\right] \sin\left(\frac{n\pi x}{L}\right)$$

■ 2. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} -x^2 + 1 & -L \le x < 0 \\ L - x & 0 \le x \le L \end{cases}$$

$$f(x) = \frac{6 + 3L - 2L^2}{12} + \frac{L}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - (2L+1)(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

$$+ \frac{1}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^3} \left[(n^2\pi^2(1 - L^2) + 2L^2)(-1)^n + n^2\pi^2(L-1) - 2L^2\right] \sin\left(\frac{n\pi x}{L}\right)$$

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■ 3. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$, assuming $L > 1$.

$$f(x) = \begin{cases} L & -L \le x < -1 \\ x & -1 \le x \le 1 \\ -L & 1 < x \le L \end{cases}$$

$$f(x) = \sum_{n=1}^{\infty} \left[ \frac{2L}{n\pi}(-1)^n - \frac{2(L+1)}{n\pi} \cos\left(\frac{n\pi}{L}\right) + \frac{2L}{n^2\pi^2} \sin\left(\frac{n\pi}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right)$$

■ 4. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} x + L & -L \le x < 0 \\ -1 & 0 \le x \le L \end{cases}$$

$$f(x) = \frac{L-2}{4} + \frac{L}{\pi^2} \sum_{n=1}^{\infty} \frac{1-(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right) + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n - 1 - L}{n} \sin\left(\frac{n\pi x}{L}\right)$$

■ 5. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$, assuming $L > 1$.

$$f(x) = \begin{cases} x^2 & |x| < 1 \\ 0 & 1 \le |x| \le L \end{cases}$$

$$f(x) = \frac{1}{3L} + \frac{2}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^3} \left[ (n^2\pi^2 - 2L^2) \sin\left(\frac{n\pi}{L}\right) + n\pi L \cos\left(\frac{n\pi}{L}\right) \right] \cos\left(\frac{n\pi x}{L}\right)$$

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■ 6. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} Lx & 0 \le x \le L \\ -x - L & -L \le x < 0 \end{cases}$$

$$f(x) = \frac{L^2 - L}{4} + \frac{L}{\pi} \sum_{n=1}^{\infty} \frac{1 - L(-1)^n}{n} \sin\left(\frac{n\pi x}{L}\right) + \frac{L(L+1)}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

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## FOURIER COSINE SERIES

■ 1. Find the Fourier cosine series representation of the function $f(x) = 3x + 3$ on $0 \leq x \leq L$.

■ 2. Find the Fourier cosine series representation of the function $f(x) = 3x^2 + 2x$ on $0 \leq x \leq L$.

■ 3. Find the Fourier cosine series representation of the function $f(x) = x^3$ function on $0 \leq x \leq \pi$.

■ 4. Find the Fourier cosine series representation of the function $f(x) = e^x$ on $0 \leq x \leq L$.

■ 5. Find the Fourier cosine series representation of the function $f(x) = x \cos x$ on $0 \leq x \leq 2\pi$.

■ 6. Find the Fourier cosine series representation of the function $f(x) = \sin(3x + 5)$ on $0 \leq x \leq \pi$.

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## FOURIER SINE SERIES

■ 1. Find the Fourier sine series representation of the function $f(x) = 2x^3 - 3x$ on $-L \leq x \leq L$.

■ 2. Find the Fourier sine series representation of the function $f(x) = -x^3$ on $-L \leq x \leq L$.

■ 3. Find the Fourier sine series representation of the function $f(x) = x^3 - 2x$ on $-L \leq x \leq L$.

■ 4. Find the Fourier sine series representation of the function $f(x) = x - x^2$ on $0 \leq x \leq L$.

■ 5. Find the Fourier sine series representation of $f(x) = x^2 - Lx$ on $0 \leq x \leq L$.

■ 6. Find the Fourier sine series representation of $f(x) = e^{L-x} + 5$ on $0 \leq x \leq L$.

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## COSINE AND SINE SERIES OF PIECEWISE FUNCTIONS

■ 1. Find the Fourier cosine series representation of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} 0 & 0 \leq x \leq \frac{L}{2} \\ x^2 & \frac{L}{2} < x \leq L \end{cases}$$

■ 2. Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq 2\pi$.

$$f(x) = \begin{cases} 2x & 0 \leq x \leq \pi \\ 4\pi - 2x & \pi < x \leq 2\pi \end{cases}$$

■ 3. Find the Fourier cosine series representation of the piecewise function on $0 \leq x \leq \pi$.

$$f(x) = \begin{cases} x^2 & 0 \leq x \leq \frac{\pi}{2} \\ x & \frac{\pi}{2} < x \leq \pi \end{cases}$$

■ 4. Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq \pi$.

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$$f(x) = \begin{cases} \sin x & 0 \leq x \leq \frac{\pi}{2} \\ \sin 2x & \frac{\pi}{2} < x \leq \pi \end{cases}$$

■ 5. Find the Fourier cosine series representation of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} 3x & 0 \leq x \leq \frac{L}{3} \\ 3x - L & \frac{L}{3} < x \leq \frac{2L}{3} \\ 3x - 2L & \frac{2L}{3} < x < L \end{cases}$$

■ 6. Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq 2$.

$$f(x) = \begin{cases} x^2 & 0 \leq x \leq 1 \\ x^2 - 1 & 1 < x \leq 2 \end{cases}$$

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